PPARG: This nuclear receptor gene activates genes that protect against **diabetes, heart disease and depression.** Its dysregulation can lead to a risk of some **cancers**

DIETS An **antioxidant diet**, rich in berries, garlic and tomatoes, negates the risk of cancer. **Plant-based food** and **rainbow diets** (vegetables of various colours) are also advised in this plan

FTO and LEPR: The FTO gene is linked with **obesity** and other **metabolic disorders**; LEPR gene gives instructions to make the leptin receptor protein that regulates body weight

DIETS Those with this predisposition can regulate calorie intake through **fasting, 1,200-calorie diets** (a low calorie plan) and **cycle syncing consumption** (adjusting eating throughout the menstrual cycle) to burn more calories

lies. "I see precision nutrition as a step forward," says Dr Suparna Mukherjee, chief nutritionist at Narayana Health City in Bengaluru. "We look into an individual's anthropometry (measurement of the physical properties of a body), biochemical markers, clinical symptoms, dietary habits and their genetic reports. We also consider their activity levels and calorie expenditure, which helps us personalise a food plan," says Dr Mukherjee. It's all about why certain foods work better for individuals based on their unique genetic and lifestyle profiles, she adds.

The right diagnostic tools can uncover details that make a difference. When 38-year-old Rohit, a fitness enthusiast, went for a genomic health assessment at Haystack Analytics in Bengaluru, he wasn't expecting to uncover anything major. The results changed his approach to health. The test flagged a heightened genetic risk for Graves' disease, an autoimmune condition that affects the thyroid.

Nutrigenomic weight loss programmes have become accessible too. When Sahil Malhotra, 41, a software engineer from Bengaluru, wanted to lose 10 kg, he opted for a personalised diet programme by a city clinic which, after procuring his genetic profile through MapMyGenome, was able to tell him why—despite trying multiple diet plans—he was unable to shed weight. "They found out that I could not digest protein too well, which is why high-protein diets failed as they left less calories for daily requirements, making me constantly hungry," he says. The results made sense, as his family was originally vegetarian.



"Nutrigenomics provides personalised nutrition grounded in molecular biology and epigenetics. While the ability to detect genetic variants is high, their clinical interpretation is an emerging area"

DR APARNA BHANUSHALI
Director, Medical Genetics, HaystackAnalytics

While more people are discovering nutrigenomics, most are adherents still of exotic biohacks like mushroom coffee and red light therapy for their nutrition needs. "Eating certain mushrooms, for instance, can have positive health benefits," says Dr Siddhant Bhargava, nutritionist, CEO and co-founder, InnerGize, a Delhi-based firm that makes mental health wearables. "Other biohacks include overloading on caffeine, or foods that improve the body's NAD (a kind of dinucleotide) levels. Nootropics (drugs like creatine and caffeine) can improve cardiac as well as brain performance." A report by Custom Market Insights, a market research firm, notes that India's health tech market—a fair share of which includes biohacking tools like wearables (such as fitness tracking bands and head-mounted displays) and brain games—is projected to touch $78.4 billion (Rs 6.7 lakh crore) by 2033.

However, the results have not always been salutary, as Rohini Bedi, 22, a marketing intern in Mumbai, discovered. On an AI-endorsed diet that involved eating in only two hours a day and weekly ice bath plunges to "naturally detox the body", Rohini ended up malnourished within a month and had to undergo clinical treatment to recover. "Individual nutrition needs and health status should take precedence over what is trending on internet search results," says Bharathi Kumar, dietitian, Fortis Hospital, Nagarbhavi, Bengaluru. Experts are also beginning to express doubts about the efficacy of biohacks. "Currently, biohacks range from theories to some proven research," says Dr Bhargava.

WORD OF CAUTION

While there's no denying the power of nutrigenomics, experts advise caution. "Biohacking diets, when done right, are incredibly empowering," says nutritionist Khushboo Jain Tibrewala. "They shift control from healthcare providers to the individual, the one actually living in the body." But she also feels that nutrigenomic biohacking, no matter how personalised, can sometimes create blind spots. This is where things like continuous glucose monitors (CGMs) come in handy. In this instance, they help people understand how their blood sugar responds to different foods. Some epigenetic drugs, on the other hand, while targeting specific diseases, can affect multiple genes, leading to undesirable side-effects on other organ systems.

Nutrigenomics and epigenetic drugs are evolving sciences, and experts like Tibrewala say that, ultimately, it is one's own state of health that should be the guide even while on a hyperpersonalised diet directed by genetic mapping. Till the time when our genes give up more of their secrets, the best data continues to be how you are feeling today. ■